- 弧微分公式

$$ds=\sqrt{(dx)^2+(dy)^2}$$

(1) 对于曲线 $y=f(x)$.

$$ds=\sqrt{(dx)^2+(dy)^2}$$

$$=\sqrt{(dx)^2+(y'dx)^2}$$

$$=\sqrt{1+(y')^2}\,dx$$

故 $\frac{ds}{dx}=\sqrt{1+(y')^2}$

(2) 参数方程 $\begin{cases} x=x(t) \\ y=y(t) \end{cases}$

$$ds=\sqrt{(dx)^2+(dy)^2}$$

$$=\sqrt{(x'(t)dt)^2+(y'(t)dt)^2}$$

$$=\sqrt{(x'(t))^2+(y'(t))^2}\,dt$$

从而 $\frac{ds}{dt} = \sqrt{(x'(t))^2 + (y'(t))^2}$

(3) 极坐标方程

$$\rho = \rho(\theta)$$

此时 $x = \rho(\theta) \cdot \cos\theta$

$y = \rho(\theta) \cdot \sin\theta$

$$ds = \sqrt{(x'(\theta))^2 + (y'(\theta))^2}\, d\theta$$

$$= \sqrt{(\rho'(\theta))^2 + (\rho(\theta))^2}\, d\theta$$

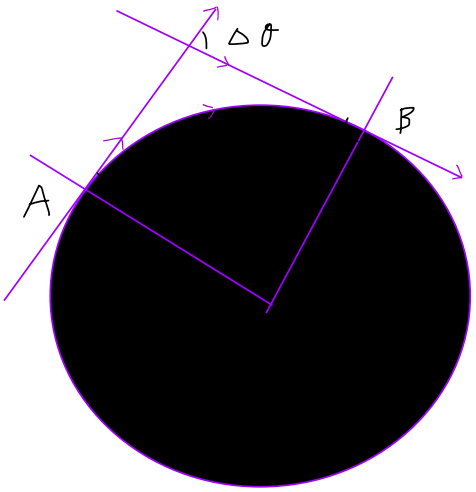
此时 $\frac{ds}{d\theta} = \sqrt{(\rho'(\theta))^2 + (\rho(\theta))^2}$

- 曲线的曲率

平均曲率 $\bar{K}=\left|\frac{\Delta\alpha}{\Delta s}\right|$

曲率 $K=\lim_{\Delta s\to 0}\left|\frac{\Delta\alpha}{\Delta s}\right|=\left|\frac{d\alpha}{ds}\right|$

例：半径为R的圆



$$\Delta s=\overset{\frown}{AB}$$

$$=R\cdot\Delta\theta$$

$$\frac{\Delta\theta}{\Delta s}=\frac{1}{R}$$

故 $K=\frac{d\theta}{ds}=\frac{1}{R}$

即圆的曲率在每点都等于 $\frac{1}{R}$.

对于圆来说，半径越大，曲率越小。从某种意义上来讲，直线可看成是半径为无穷大的圆。

- 曲率的计算。

(1) 若 $y=f(x)$ 二阶可导，则

$$\tan\alpha = y', \quad \alpha = \arctan y'$$

$$d\alpha = \frac{y''}{1+(y')^2}dx$$

又 $ds = \sqrt{1+(y')^2}\,dx$

故 $$k = \left|\frac{d\alpha}{ds}\right| = \frac{|y''|}{(1+(y')^2)^{\frac{3}{2}}}$$

(2) 若曲线为 $\begin{cases} x=\varphi(t) \\ y=\psi(t) \end{cases}$, 则

$$k=\frac{\left|\varphi'(t)\psi''(t)-\varphi''(t)\cdot\psi'(t)\right|}{\left((\varphi'(t))^2+(\psi'(t))^2\right)^{\frac{3}{2}}}$$

- 例: 求摆线

$$\begin{cases} x=a(\theta-\sin\theta) \\ y=a(1-\cos\theta) \end{cases} \quad (0\le\theta\le 2\pi)$$

的曲率, 并讨论在摆线上哪一点曲率最小? 并求最小曲率.

解:

$$x'(\theta)=a(1-\cos\theta)$$

$$y'(\theta)=a\sin\theta$$

$$x''(\theta)=a\sin\theta$$

$$y''(\theta)=a\cos\theta$$

$$k=\frac{\left|x'(\theta)y''(\theta)-x''(\theta)\cdot y'(\theta)\right|}{\left((x'(\theta))^2+(y'(\theta))^2\right)^{\frac{3}{2}}}$$

$$=\frac{a^2(1-\cos\theta)}{\left(a^2(2-2\cos\theta)\right)^{\frac{3}{2}}}$$

$$=\frac{1}{2a\cdot(2-2\cos\theta)^{\frac{1}{2}}}$$

$$=\frac{1}{2a\cdot 2\sin\frac{\theta}{2}}=\frac{1}{4a\cdot\sin\frac{\theta}{2}}$$

$(0\le\theta\le 2\pi)$

故当 $\sin\frac{\theta}{2}=1$ 时，即 $\theta=\pi$ 时，曲率 k 最小，此时 $k=\frac{1}{4a}$

$(x,y)=(a\pi,2a)$.

- 例：求心脏线 $\rho = a(1+\cos\theta)$ 的曲率半径. $(a>0)$

解：$x = \rho\cos\theta = a(\cos\theta + \cos^2\theta)$

$y = \rho\sin\theta = a(\sin\theta + \sin\theta\cdot\cos\theta)$

$$k = \frac{|x'(\theta)y''(\theta) - x''(\theta)\cdot y'(\theta)|}{\left((x'(\theta))^2 + (y'(\theta))^2\right)^{\frac{3}{2}}}$$

（计算留作习题）.

- 例：求曲线 $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ 的曲率. $(a>0)$

解：令 $x^{\frac{1}{3}} = u$, $y^{\frac{1}{3}} = v$. $a^{\frac{1}{3}} = r$

则 $u^2 + v^2 = r^2$

可设 $\begin{cases} u = r\cos\theta \\ v = r\sin\theta \end{cases}$

于是 $\begin{cases} x = a\cos^3\theta \\ y = a\sin^3\theta \end{cases}$

利用

$$k = \frac{|x'(\theta)\cdot y''(\theta) - x''(\theta)\cdot y'(\theta)|}{\left((x'(\theta))^2 + (y'(\theta))^2\right)^{\frac{1}{2}}}$$

求曲率.

(留作习题)

- 曲率圆

半径 $\rho=\frac{1}{k}$

圆心过法线（指向凹侧）

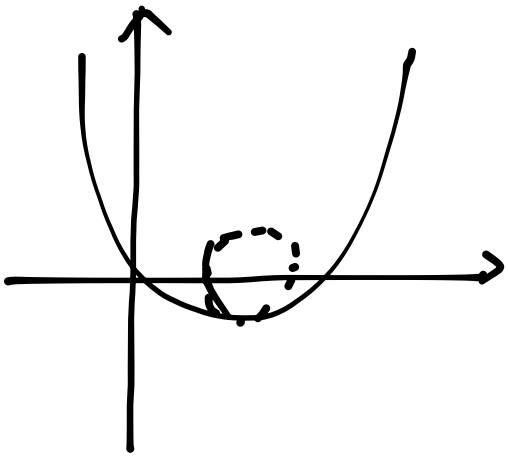
例：求抛物线 $y=x^2-4x+3$ 在顶点处的曲率圆.

解：$y=x^2-4x+3=(x-2)^2-1$

顶点为 $(2,-1)$.

在顶点处切线水平

（实际上 $y'\big|_{x=2}=0$）



$$k=\frac{|y''|}{(1+(y')^2)^{\frac{3}{2}}}\Bigg|_{x=2}=\frac{2}{(1+(2x-4)^2)^{\frac{3}{2}}}\Bigg|_{x=2}$$

$=2$

曲线圆半径 $R=\frac{1}{k}=\frac{1}{2}$

曲率圆圆心为 $(2,-1+\frac{1}{2})=(2,-\frac{1}{2})$

曲率圆方程为

$$(x-2)^2+(y+\frac{1}{2})^2=\frac{1}{4}$$

作业：

习题3-6 A类

1. (1)

2. (3)

3. (1)

4. (1)